

Causal inference with uncertain record linkage: a finite-population research compendium

Research draft

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Abstract

A linkage error rate does not determine causal bias. The fields moved by a false link, which units are linked correctly, exclusion from the analysis, and the treatment assignment mechanism all matter. We formalize these distinctions, prove finite-population results, and implement two remedies: validation-sample correction and inference over feasible reconstructions. The correction permits outcome-dependent linkage when an independent probability sample validates the necessary fields. Graph ranges require separate assumptions about candidate recall, overlap, and unmatched outcomes; ranges of estimates alone are not causal confidence intervals. Exhaustive small cases check the derivations, and simulations illustrate scope and failure. A read-only application reconstructs the Rajasthan reservation–MNREGA linkage pipeline. The broad problem and several remedies already have substantial precedents. This draft makes no claim that the results establish a novel statistical method.

1 What existing work establishes

Causal inference with uncertain identities is an existing field. Propensity-score subclassification methods already trade linkage confidence against downstream estimation error [1]. Bayesian approaches jointly infer identity and causal quantities [2], including covariates distributed across files [3]. Treatment-roster linkage has been analyzed with multiple imputation and an explicit sensitivity analysis for informative linkage [4]. Experimental linking errors can reduce statistical power [5].

The closest broad comparator treats each of X , treatment, and outcome as the field separated from the other two, derives naive-estimator bias, and develops corrections with asymptotic inference [6]. Its mismatch and auxiliary-variable assumptions are conditional on covariates, not a blanket claim that all match quality is unrelated to potential outcomes. Other methods allow analysis variables to inform linkage. Consequently, “existing work only keeps high-precision matches” would misdescribe the literature. These are statements about specified methods; no systematic survey of applied practice was conducted here.

Optimization over feasible identity resolutions already bounds aggregate queries [7]. Optimization over acceptable treated–control pairings also exists [8], although those pairings join different people, not records for the same person. Validation-based correction belongs to the established two-phase estimation literature [9]. The contribution question here is narrower: whether retaining the candidate graph, assignment restrictions, and joint field structure yields a useful causal analysis beyond these methods. That question remains open. The literature matrix records assumptions and exact source locations; absence from this matrix is not proof of novelty.

2 Target, files, and sources of randomness

Fix N target units $i = 1, \dots, N$, baseline covariates X_i , and potential outcomes $Y_i(0), Y_i(1)$. Initially assume no interference, consistency, and perfect compliance. Assignment Z is uniform over all vectors with m treated units, $0 < m < N$, independent of the fixed potential outcomes. Write $c = N - m$ and

$$Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0), \quad \tau_N = N^{-1} \sum_i \{Y_i(1) - Y_i(0)\}.$$

The oracle difference in means is

$$\hat{\tau}_O = \sum_i w_i(Z) Y_i, \quad w_i(Z) = Z_i/m - (1 - Z_i)/c.$$

Its assignment expectation is τ_N and its variance is $S_1^2/m + S_0^2/c - S_\tau^2/N$, where each S^2 uses divisor $N - 1$. The Neyman variance estimator omits the unidentifiable negative term. Its expectation is conservative; this does not make a finite-sample normal interval exact.

File inclusion indicators R_i^A, R_i^B determine which units have records. Noisy identifiers Q^A, Q^B generate a bipartite candidate graph G . A true partial mapping L^* links records for the same unit; it is injective only if each file has at most one record per relevant unit and period. Multiple villages in a panchayat, boundary changes, and repeated measurements must be resolved before imposing this assumption. An estimated linkage $\hat{L}$ and retention rule S produce the analysis data. The true link can be absent from G even when a source has several candidates.

A reconstruction $D(L)$ moves all fields recorded together. Three useful layouts are $(X, Z) \mid Y$, $(X, Y) \mid Z$, and $X \mid (Z, Y)$. Separately uncertain covariates and outcomes require separate maps with explicit consistency restrictions. Reassigning individual columns independently can create impossible records. We distinguish absent counterparts, omitted true candidate edges, wrong accepted edges, ambiguous candidates, donor reuse, and source exclusions. All may depend on X , assignment, realized outcomes, or latent potential outcomes unless a result rules that dependence out.

Expectations must specify their randomness: assignment, population sampling, identifier corruption, validation sampling, or an algorithm's random choices. Sections 3–5 use fixed populations and state when averaging over validation or linkage as well. The simulation regenerates populations, then assignments and errors. Conditioning on a chosen linkage answers a different uncertainty question from repeating the entire process.

With noncompliance, distinguish assignment Z , treatment receipt $D(Z)$, and outcomes $Y(D, Z)$. The default target is the intention-to-treat effect. A ratio of outcome and receipt contrasts targets a complier effect only with additional exclusion, monotonicity, relevance, and design assumptions. Linkage uncertainty in both numerator and denominator must be propagated jointly. A near-zero feasible first stage can make a ratio set unbounded; dividing marginal endpoint bounds is generally invalid. Reservation status and the sex of the elected officeholder are therefore not interchangeable treatments.

3 How bias and variance enter

Proposition 1 (Bookkeeping identities, established). *For any two square-integrable estimators on the same probability space, define $\Delta = \hat{\tau}_L - \hat{\tau}_O$. Then*

$$\begin{aligned} \mathbb{E} \hat{\tau}_L - \tau_N &= (\mathbb{E} \hat{\tau}_O - \tau_N) + \mathbb{E} \Delta, \\ \text{Var}(\hat{\tau}_L) &= \text{Var}(\hat{\tau}_O) + \text{Var}(\Delta) + 2 \text{Cov}(\hat{\tau}_O, \Delta). \end{aligned}$$

Proof. Substitute $\hat{\tau}_L = \hat{\tau}_O + \Delta$, and use linearity of expectation and the variance of a sum. No independence is needed. $\square$

The covariance cannot be discarded. A procedure returning zero under a zero-effect population can have less variance than the oracle. A smaller reported standard error can coexist with greater bias. For a quadratic-loss decomposition, $\text{MSE}_L = \text{Var}(\hat{\tau}_L) + (\mathbb{E}\hat{\tau}_L - \tau_N)^2$. A bootstrap of fixed linked rows neither discovers omitted true links nor automatically accounts for the negative dependence created by donor exclusivity.

To separate mechanisms, let $a_O(S), r_O(S)$ be clean adjusted and unadjusted estimators on a selected set, and $a_L(S), r_L(S)$ their reconstructed versions. Then

$$a_L(S) - a_O(N) = \underbrace{a_O(S) - a_O(N)}_{\text{selection}} + \underbrace{r_L(S) - r_O(S)}_{\text{reconstruction}} + \underbrace{[a_L(S) - r_L(S)] - [a_O(S) - r_O(S)]}_{\text{adjustment distortion}}.$$

This identity requires all estimators to exist. Empty arms or failed fits are failures, not zeros. A different order gives a different allocation of interactions; these are not uniquely identified causal components of linkage error.

Proposition 2 (Pre-assignment selection). *For a fixed retained subset S , conditional on its containing both assignment arms, its unadjusted contrast has expectation $\tau_S = |S|^{-1} \sum_{i \in S} \tau_i$. Writing $s_i = 1\{i \in S\}$ and $\bar{s} = |S|/N$, $\tau_S - \tau_N = \text{Cov}_N(s_i, \tau_i)/\bar{s}$, where Cov_N uses divisor N .*

Proof. Conditional on the treated count within S , all subsets of that size are equally likely. Each arm mean is unbiased for its corresponding potential-outcome mean in S . Averaging over nonempty counts proves the first assertion. Expand the covariance for the second. $\square$

Thus perfect precision does not remove target-population bias. If S depends on realized assignment or outcomes, this proof fails and even the retained-population contrast need not be causally identified. Inclusion weighting needs positive, identified inclusion probabilities and the appropriate selection ignorability; it cannot recover units that have no chance of being observed without extrapolation.

Proposition 3 (Exact finite-population permutation expectation). *Suppose outcome linkage is a fixed permutation π , independent of assignment. Set $F_i = 1\{\pi(i) = i\}$ and $q = N^{-1} \sum_i F_i$. Then*

$$\begin{aligned} \mathbb{E}_Z \sum_i w_i(Z) Y_{\pi(i)} &= \frac{\sum_i F_i \tau_i - \tau_N}{N - 1} \\ &= \frac{Nq - 1}{N - 1} \tau_N + \frac{N}{N - 1} \text{Cov}_N(F_i, \tau_i). \end{aligned}$$

Proof. For $i = j$, $\mathbb{E}\{w_i Y_i\} = \tau_i/N$. For $i \neq j$, $\mathbb{E}w_i = 0$ and $\mathbb{E}(w_i Z_j) = -1/[N(N - 1)]$, obtained from $\Pr(Z_i = Z_j = 1) = m(m - 1)/[N(N - 1)]$. Therefore $\mathbb{E}(w_i Y_j) = -\tau_j/[N(N - 1)]$. Each outcome appears exactly once. Summing diagonal and off-diagonal terms proves the first line; expanding the covariance proves the second. This holds for unequal arm sizes as well. $\square$

For random permutations independent of assignment, average the formula over their law; F_i is replaced by its correctness probability in the first line. Homogeneous effects or equal correctness probabilities give a scalar attenuation factor. Without either restriction, average precision alone is insufficient. A fixed derangement gives $-\tau_N/(N - 1)$, a small sign reversal rather than literal zero. This finite-population calculation is a derivation supplied here, with no priority claim.

Example 1 (Same precision, different causal information). *A within-arm permutation preserves the realized difference in means exactly, however many identities are wrong. Swapping entire treatment and control outcome vectors in an equally allocated experiment reverses its sign. Both can have zero correct links. The first map depends on assignment, so applying the preceding independent-permutation formula to it would be an error. Moving Z and Y jointly also preserves their unadjusted contrast; moving only X can still damage observational adjustment.*

For outcome contamination let e_z be the wrong-link fraction in observed arm z , μ_z its correctly linked outcome mean, and ν_z the corresponding donor mean. The linked contrast equals $(1 - e_1)\mu_1 + e_1\nu_1 - (1 - e_0)\mu_0 - e_0\nu_0$. This is a mixture identity, not an identification theorem. Attenuation requires restrictions on the donor means. When $\mu_1 = 1, \mu_0 = 0, e_1 = e_0 = .1$, choosing $\nu_1 = 10, \nu_0 = 0$ gives 1.9; choosing $\nu_1 = -20$ gives -1.1 . False links can amplify or reverse an effect.

For a different model, suppose treatment is randomized with probability p and misclassified nondifferentially: $a = \Pr(\tilde{Z} = 0 \mid Z = 1)$, $b = \Pr(\tilde{Z} = 1 \mid Z = 0)$, with label error independent of potential outcomes conditional on Z . Bayes' rule gives

$$\mathbb{E}[Y \mid \tilde{Z} = 1] - \mathbb{E}[Y \mid \tilde{Z} = 0] = \left\{ \frac{p(1-a)}{p(1-a) + (1-p)b} - \frac{pa}{pa + (1-p)(1-b)} \right\} \tau.$$

The proof substitutes $\mathbb{E}[Y \mid Z = z] = \mathbb{E}[Y(z)]$ into each posterior mixture. A roster with fixed treated count induces dependencies not described by independent Bernoulli misclassification. It needs the full assignment or linkage model.

4 Validation as a remedy

The following is a standard two-phase difference-estimation construction specialized to a completely randomized experiment. It is useful because it does not require an ignorable linkage mechanism.

Proposition 4 (Validation correction and conservative variance expectation). *Let h_i be any fully observed linked score, possibly constructed using all linked records and assignments, with $H = \sum_i h_i$. Let an independent simple random sample A of $k \geq 2$ anchor units validate both Y_i and Z_i , after all h_i are fixed. Define $d_i = w_i(Z)Y_i - h_i$. Then*

$$\hat{\tau}_V = H + \frac{N}{k} \sum_{i \in A} d_i, \quad \mathbb{E}_A(\hat{\tau}_V \mid Z, \text{files}) = \hat{\tau}_O.$$

If both assignment arms have at least two units,

$$\hat{V}_V = \hat{V}_{O,A} + N^2(1 - k/N)s_{d,A}^2/k$$

has expectation $\text{Var}_{Z,A}(\hat{\tau}_V) + S_\tau^2/N$, conditional on the finite population and averaging over any linkage randomness, where

$$\hat{V}_{O,A} = \frac{1}{\pi_2} \left\{ \frac{\sum_{i < j \in A: Z_i = Z_j = 1} (Y_i - Y_j)^2}{m^2(m-1)} + \frac{\sum_{i < j \in A: Z_i = Z_j = 0} (Y_i - Y_j)^2}{c^2(c-1)} \right\}, \quad \pi_2 = \frac{k(k-1)}{N(N-1)}.$$

Proof. The validation sample mean of d is unbiased for its population mean. Its conditional variance is $(1 - k/N)S_{d,A}^2/k$, and $s_{d,A}^2$ is unbiased for $S_{d,A}^2$. The conditional mean of the validation error is zero, so its covariance with the oracle estimator is zero. The law of total variance gives the result once the oracle variance is estimated. Each pair has inclusion probability π_2 . Using $\sum_{i < j} (Y_i - Y_j)^2 = n \sum_i (Y_i - \bar{Y})^2$ within each arm, the expectation of $\hat{V}_{O,A}$ over validation equals the full-data Neyman variance estimator. Its assignment expectation exceeds the oracle variance by S_τ^2/N . $\square$

This result allows errors in treatment as well as outcomes, and arbitrary dependence of linkage on either. It requires a complete anchor population, known assignment counts, error-free validation of the necessary fields, and a probability validation sample independent of the fixed scores. Reviewing only doubtful links violates this sampling assumption. Scores can be zero for unlinked units, but validation must still be capable of recovering their true outcomes. If records are intrinsically unavailable, that requirement is not met. Targeted review needs its own inclusion probabilities; the implemented formula cannot be reused unchanged. Normal intervals are assessed by simulation, not guaranteed by the variance result.

A model-based alternative divides a linked contrast by $\hat{\kappa} = (N\hat{q} - 1)/(N - 1)$. Even when the model justifies the population factor, estimating and inverting it creates finite-sample bias. The delta variance is

$$V(\hat{T}/\hat{\kappa}) \simeq V(\hat{T})/\kappa^2 + T^2 V(\hat{\kappa})/\kappa^4 - 2T \text{Cov}(\hat{T}, \hat{\kappa})/\kappa^3.$$

With fixed permutation and an independent validation sample, the covariance term is zero under assignment-validation averaging. A random permutation or shared fitted linkage model can invalidate that simplification. The simulation comparator uses a working linked-data Neyman variance and is labeled approximate. Near-zero factors are reported as failures. Fitting a mixture or joint Bayesian model is another established route; the present code does not reproduce every estimator in the literature matrix.

5 Feasible reconstructions and causal uncertainty

For each source i and donor j , let $x_{ij} \in \{0, 1\}$ indicate a link and let $u_i \in \{0, 1\}$ indicate an outside option. The implemented feasible set imposes

$$\sum_j x_{ij} + u_i = 1, \quad \sum_i x_{ij} \leq 1, \quad K_{\min} \leq \sum_{ij} x_{ij} \leq K_{\max}, \quad \sum_{(i,j) \notin G} x_{ij} \leq B_G.$$

Only explicitly permitted sources may take the outside option. Accepted “certain” links can be imposed with a budget B_C for violations. These are sensitivity parameters, not probabilities estimated from a score margin. A missing candidate edge and a genuinely absent outcome are different events. An off-graph budget permits known donors outside the graph; an outside option requires outcome support for an unobserved or unmatched outcome. Both may be needed.

Proposition 5 (Sharp ranges of a linear reconstructed statistic). *For fixed weights w_i , donor outcomes y_j , and independently unconstrained missing outcomes $v_i \in [a, b]$, the sharp lower and upper values of $\sum_{ij} w_i y_j x_{ij} + \sum_i w_i v_i u_i$ over the stated feasible set are binary linear programs. Outside coefficients are $\min(w_i a, w_i b)$ for the lower problem and $\max(w_i a, w_i b)$ for the upper problem.*

Proof. For any fixed matching, the outside terms minimize or maximize separately at the appropriate endpoints. The remaining objective is linear in binary assignment variables. Every feasible reconstruction corresponds to one feasible vector of these variables, and each feasible vector constructs a permitted reconstruction with the chosen endpoints. The finite optimum is attained, proving sharpness relative to these assumptions. Extra joint restrictions on missing outcomes could narrow the range. $\square$

Without budgets or other non-network constraints this reduces to an assignment/min-cost flow problem. General budgets can require integer programming. “Sharp” here describes the range of a statistic over specified reconstructions. It does not establish sharp partial identification of τ_N , because unobserved potential outcomes remain.

For m treated-roster records that must match injectively into a complete population of N outcome records, all units outside the roster image are controls. Therefore

$$\hat{\tau}_L = \frac{N}{m(N-m)} \sum_{j \in \text{image}(L)} y_j - \frac{\sum_j y_j}{N-m}.$$

The denominators remain fixed. Incomplete roster coverage breaks this expression. Ordinary least-squares coefficients with a fixed design matrix are also linear in outcomes; moving treatment or covariates changes the design matrix, and generally requires optimizing the entire estimator rather than holding its weights fixed.

Example 2 (Outside options and global influence). *Two sources both have candidates with outcomes 0 and 10. Requiring both to match gives mean 5. Allowing either to be unmatched with support $[0, 10]$ gives mean range $[0, 10]$. Presence of candidates does not justify the first cardinality restriction. Next, let source 1 have candidates $b_1 = 0, b_2 = 0$, and source 2 have $b_1 = 0, b_3 = 100$, with both required to match. The mean range is $[0, 50]$. Learning that source 1 uses b_1 collapses the mean to 50; learning it uses b_2 leaves the range unchanged. A zero local outcome range can thus carry review value because of competition for donors. ‘review_priority()’ reports best and guaranteed reductions, without assigning probabilities to possible reviewer answers.*

Proposition 6 (Coverage and graph failure are separate). *Suppose a full-data confidence set $C(D^*)$ covers a fixed causal target with probability at least $1 - \alpha$, and the random reconstruction set $\mathcal{D}$ contains D^* with probability at least $1 - \delta$. Then $\bigcup_{D \in \mathcal{D}} C(D)$ covers with probability at least $1 - \alpha - \delta$. For valid full-data sharp-null p -values, $p_{\max} = \sup_{D \in \mathcal{D}} p(D)$ rejects at level α with probability at most $\alpha + \delta$.*

Proof. Failure can occur only if the true reconstruction is excluded or the full-data procedure fails at the truth. Apply the union bound. On the containment event, $p_{\max} \geq p(D^*)$, proving the test statement. $\square$

No independence between graph construction and the experiment is needed for this unconditional argument. It does require that $C(D^*)$ or $p(D^*)$ be a valid oracle procedure on the full-data probability space. If an analyst freezes an outcome-dependent selected graph and permutes only the selected rows using a fictional assignment law, that oracle premise is generally false. Either reproduce the selection/linkage process under each assignment or derive the correct conditional law. Our exact test enumerates the original complete assignment support, with all anchors retained, for every feasible full linkage. It tests the sharp zero-effect null; it is not a test of the weak zero-ATE null under heterogeneous effects.

Proposition 7 (An explicit conservative causal confidence set). *Suppose every potential outcome lies in a known interval $[a, b]$, all anchors are retained, and assignment is complete with counts m, c . Let $[L, U]$ bound the oracle realized contrast over feasible reconstructions, and set*

$$r = (b - a) \sqrt{\log(4/\alpha)/2} \{m^{-1/2} + c^{-1/2}\}.$$

Then $[L - r, U + r] \cap [-(b - a), b - a]$ covers τ_N with probability at least $1 - \alpha - \delta$, where δ bounds graph/reconstruction failure.

Proof. Each observed arm mean is a simple-random-sample mean of its fixed potential-outcome population. Hoeffding’s bound for sampling without replacement [10] gives, for each arm, absolute deviation at most $(b - a) \sqrt{\log(4/\alpha)/(2n_z)}$ except on probability $\alpha/2$. The triangle inequality and union bound control the difference. Apply the previous proposition and intersect with the logical ATE support. $\square$

These intervals can be wide; that is the cost of weak assumptions. The observed sample minimum and maximum cannot substitute for known potential-outcome support. More powerful design-specific intervals can be unioned, but their uniform validity must be established separately. A credible interval from a linkage model answers a different question and requires the specified posterior model.

6 Observational extension and limits of identification

With observational assignment, even perfect identity does not identify an ATE without causal assumptions. Under consistency, conditional exchangeability $\{Y(0), Y(1)\} \perp Z \mid X$, and positivity, an oracle estimator can use $e(X) = \Pr(Z = 1 \mid X)$ and outcome regressions. If the linked covariates are $\tilde{X}$, exchangeability given X does not imply exchangeability given $\tilde{X}$. A linkage model must preserve the conditioning information or quantify its loss. Standard doubly robust estimation protects against misspecification of one nuisance model on correctly measured data; it does not automatically protect against incorrect identities shared by both nuisance fits.

Proposition 8 (Correct measured-data models need not remove confounding). *Let X be Bernoulli with probability $1/2$. Set $\Pr(Z = 1 \mid X) = .2 + .6X$, $Y(0) = X$, and $Y(1) = X + 1$. Observe $W = X$ with probability q and an independent Bernoulli($1/2$) draw otherwise, with this mechanism independent of (X, Z) apart from the copying. Even using the true measured-covariate propensity $e_W(w) = .5 + .6q(w - .5)$, the expectation of its inverse-propensity estimator is*

$$1 + \frac{.6(1 - q^2)}{1 - .36q^2},$$

whereas the causal effect and oracle X -adjusted expectation both equal one.

Proof. Bayes' rule gives $\Pr(X = 1 \mid W = w) = .5 + q(w - .5)$, hence $\text{Var}(X \mid W) = (1 - q^2)/4$. Within either W stratum, $\text{Cov}(X, Z \mid W) = .6 \text{Var}(X \mid W)$ and $e_W(1 - e_W) = (1 - .36q^2)/4$. The conditional difference in mean baseline outcomes is the ratio of these last two quantities. Adding the treatment effect and averaging over W proves the expression. The same observed-data contrast is obtained with correctly specified measured-data outcome regressions, so double robustness does not repair the lost exchangeability. $\square$

This replacement model describes an external donor distribution, not a fixed one-to-one permutation. The separate script `scripts/observational.R` checks the formula at five values of q with known propensities. Here q is a copying probability; an incorrect donor can coincidentally have the same binary covariate, so q is not observed covariate agreement. This is an illustration of residual confounding, not a novelty claim or a general observational remedy.

The bookkeeping identity still holds, but its oracle-bias term can be nonzero. The validation difference principle applies to known-propensity oracle scores, with the appropriate sampling variance. Estimated propensity scores and outcome models need joint influence-function, cross-fitting, or resampling calculations that include validation and linkage estimation. The finite-randomization variance formula above cannot simply be copied. Likewise, a union over linkages inherits causal coverage only from a valid observational oracle procedure. Linkage sensitivity does not bound unmeasured confounding.

Proposition 9 (Uninformative identities can prevent causal identification). *Consider two anchor units with exactly one assigned treatment. File A records their identities and assignments. File B records the unordered pair of realized outcomes, with no informative identifier; its candidate graph is*

complete. Even though every unit appears once and assignment is randomized, the joint distribution of these observations does not identify τ_2 .

Proof. In population P, both units have $Y_i(0) = 0, Y_i(1) = 1$, so $\tau_2 = 1$. In population Q, both have $Y_i(0) = 1, Y_i(1) = 0$, so $\tau_2 = -1$. Under either population and either possible assignment, the unordered outcome file is exactly $\{0, 1\}$. File A's two possible assignment vectors each have probability $1/2$ under both populations. Identifiers and the candidate graph are the same. Thus the entire observable law is identical while the causal target differs. $\square$

The unordered-file condition can equivalently be represented by sorting outcome rows or randomly ordering them independently of identity. In that representation the correct row mapping need not be independent of assignment, which is permitted here but was excluded in the fixed-permutation theorem. This example proves more than two compatible realized contrasts: it constructs identical observable probability laws. Candidate containment and full overlap alone do not identify the effect. Validation, informative identifiers, or restrictions on the linkage mechanism add information. Randomization still supports conservative set inference with the known outcome support.

7 Simulation and empirical illustration

The executable protocol in `research/simulation-design.md` specifies populations, corruption mechanisms, methods, seeds, and reporting before the pilot. The main simulation crosses field layout and linkage mechanism, includes unequal treatment allocation and heterogeneous effects, and separates controlled candidate graphs from synthetic identifier corruption. The primary target stays the full finite population; selected-target shifts are reported separately. Normal-approximation coverage is measured rather than assumed. The graph confidence sets use the preceding bounded-outcome result, and their widths should be judged alongside their protection. Under independent outcome linkage, the naive estimator has bias -0.304 ; the validation difference estimator has bias -0.004 (MCSE 0.007). Under selection, their biases are -0.673 and -0.018 . Validation restores the fixed-population target under its sampling assumptions; the normal approximation still needs separate assessment.

The graph causal sets are protective but weak in these designs. For independent outcome linkage their mean width is 7.00 and power is 0.000 ; validation intervals have width 1.28 and power 0.860 . The identifier design permits every source an outside option, so its graph sets can be logically uninformative. The separate estimator ranges still measure how much one-to-one coupling constrains reconstruction. This experiment does not establish a competitive general-purpose graph interval.

The Rajasthan application is implemented independently in `application/run.R`. It hashes source inputs, reproduces the current accepted links and regression specification, preserves all candidates before nearest-neighbor selection, and compares exact and fuzzy linkage on both method-specific and common cohorts. Its source audit separates reservation assignment from officeholder sex and records the missing historical allocation information. The application is a linkage and estimator sensitivity exercise; it does not identify actual linkage bias without independently validated identities. General legal provisions for lotteries do not establish the year-specific randomization support needed for an exact test. On the fixed 4,355-GP fuzzy cohort, candidate assignment extrema for the 2005 coefficient are $[-29.72, 31.89]$, compared with $[-48.60, 47.27]$ when donor reuse is allowed. These are conditional estimator ranges, not causal confidence intervals.

Table 1: Finite-population ATE estimation with uncertain linkage. Each cell uses independent simulated populations of 80 units; validation methods audit 40. Bias is in outcome units. Coverage is for approximate 95% Wald intervals; MCSE is Monte Carlo standard error.

Scenario	Method	Bias	Coverage	MCSE
independent y	naive	-0.304	0.795	0.009
independent y	attenuation	0.003	0.959	0.004
independent y	audit difference	-0.004	0.945	0.005
covariate y	naive	-0.471	0.574	0.011
covariate y	attenuation	-0.237	0.893	0.007
covariate y	audit difference	-0.002	0.940	0.005
assignment y	naive	-0.005	0.954	0.005
assignment y	attenuation	0.436	0.831	0.008
assignment y	audit difference	0.005	0.948	0.005
independent t	naive	-0.300	0.809	0.009
independent t	attenuation	0.013	0.954	0.005
independent t	audit difference	0.015	0.942	0.005
selection	naive	-0.673	0.339	0.011
selection	attenuation	-0.526	0.745	0.010
selection	audit difference	-0.018	0.943	0.005
identifiers	naive	-0.122	0.921	0.006
identifiers	attenuation	-0.000	0.954	0.005
identifiers	audit difference	-0.003	0.942	0.005

Table 2: Point intervals and graph confidence sets, with 2000 replications per cell. Width is in outcome units. Null rejection uses sharp-zero effect runs; power uses positive-effect runs. Coverage refers to the finite-population ATE. Graph sets use known potential-outcome support [0,6].

Scenario	Method	Coverage	Width	Null rejection	Power
identifiers	audit difference	0.942	1.12	0.055	0.942
identifiers	graph	1.000	12.00	0.000	0.000
identifiers	naive	0.921	1.01	0.057	0.925
independent y	audit difference	0.945	1.28	0.053	0.860
independent y	graph	1.000	7.00	0.000	0.000
independent y	naive	0.795	1.04	0.057	0.751
selection	audit difference	0.943	2.29	0.057	0.385
selection	graph	1.000	11.26	0.000	0.000
selection	naive	0.339	1.10	0.047	0.222

Table 3: Rajasthan completed projects, 2011–2014: reproduced 2005 reservation coefficient, controlling for 2010 reservation. Standard errors condition on linked data. Cohort changes differ from linkage reconstruction uncertainty.

Link rule	Cohort	GPs	Estimate	OLS SE
fuzzy	method	4355	-0.026	1.880
fuzzy	common	1861	-2.992	2.902
exact	method	2036	-2.994	2.740
exact	common	1861	-2.992	2.902

8 Result status and next empirical requirements

The bias/variance decomposition, selection weighting, mixture calculations, two-phase correction, graph optimization, and union-of-confidence-sets argument use established tools. The exact finite-population permutation and validation-variance formulas are proved and tested here, but their priority is unresolved. The globally coupled review example is a useful diagnostic, not by itself a paper contribution.

The implemented remedies require different information. A probability validation sample buys design-based correction without outcome-independent linking. A credible candidate set and outcome support buy sensitivity intervals without a calibrated linkage probability model. An informative joint model can provide narrower estimates at the cost of modeling assumptions. The application needs independent identity checks, historical assignment rosters and lottery rules, and a dated geographic crosswalk before it can support stronger causal or linkage-bias claims. Comparing all contemporary model-based estimators, extending valid inference to observational nuisance estimation, and developing efficient sharp-null optimization for large graphs remain research extensions rather than completed claims.

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